# EXPERIMENT NO. 4: ONLINE TRAIN RESERVATION SYSTEM

## AIM:

To implement an online train reservation web system supporting train searches by origin/destination stations, class-wise seat quotas, PNR generation, real-time PNR status tracking, and automated refund percentage calculation upon cancellation.

## PSEUDOCODE (IN 6 SIMPLE POINTS):

**Step 1:** Authenticate passenger credentials and accept journey criteria: Origin, Destination, Date, and Class (1A, 2A, 3A, SL).

**Step 2:** Search and display list of operating trains with departure/arrival timings and real-time quota status.

**Step 3:** Accept passenger details (name, age, berth preference) and calculate total train fare including GST.

**Step 4:** Generate an authenticated 10-digit PNR number upon payment simulation and mark booking status as Confirmed or RAC.

**Step 5:** Provide a PNR inquiry search tool to retrieve passenger berth and coach allocation details.

**Step 6:** Execute ticket cancellation logic by computing refund deductions based on time remaining before departure.

## SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>RailYatra - Train Reservation System</title>
  <style>
    body { font-family: 'Segoe UI', sans-serif; margin: 0; background: #eef2f6; }
    .navbar { background: #003366; color: white; padding: 15px 30px; display: flex; justify-content: space-between; }
    .main { max-width: 950px; margin: 25px auto; background: white; padding: 25px; border-radius: 8px; box-shadow: 0
4px 12px rgba(0,0,0,0.08); }
    .grid-inputs { display: grid; grid-template-columns: repeat(4, 1fr); gap: 12px; margin-bottom: 20px; }
    input, select, button { padding: 10px; border: 1px solid #b0bec5; border-radius: 4px; box-sizing: border-box; width:
100%; }
    button { background: #ff6600; color: white; border: none; font-weight: bold; cursor: pointer; }
    button:hover { background: #e65c00; }
    .train-item { border: 1px solid #cfd8dc; padding: 15px; border-radius: 6px; margin-bottom: 12px; display: flex; justify-
content: space-between; align-items: center; }
    .badge-avail { background: #e8f5e9; color: #2e7d32; padding: 4px 10px; border-radius: 4px; font-weight: bold; }
    .pnr-box { background: #f0f7ff; border: 1px solid #bbdefb; padding: 18px; border-radius: 8px; margin-top: 20px; }
  </style>
</head>
<body>
  <div class="navbar">
    <h2>National Rail Reservation Portal</h2>
    <span>IRCTC Web Simulator</span>
  </div>
  <div class="main">
    <h3>Search Trains Between Stations</h3>
    <div class="grid-inputs">
      <input type="text" id="srcStation" placeholder="Origin Station (e.g., MAS)">
      <input type="text" id="destStation" placeholder="Destination (e.g., SBC)">
      <input type="date" id="jDate">
      <select id="tClass">
        <option value="Sleeper (SL) - Rs. 320">Sleeper (SL) - Rs. 320</option>
        <option value="AC 3 Tier (3A) - Rs. 840">AC 3 Tier (3A) - Rs. 840</option>
        <option value="AC 2 Tier (2A) - Rs. 1250">AC 2 Tier (2A) - Rs. 1250</option>
      </select>
    </div>
```

```
      <button onclick="searchTrains()">Find Available Trains</button>

      <div id="trainList" style="margin-top: 20px; display: none;">
        <div class="train-item">
          <div>
            <h4>12607 - Lalbagh Express (MAS &rarr; SBC)</h4>
            <p>Dep: 06:00 AM | Arr: 11:30 AM | Runs: Daily</p>
          </div>
          <div>
            <span class="badge-avail">Seats: Available - 42</span><br><br>
            <button onclick="openBooking('12607 Lalbagh SF Exp')" style="width: auto; padding: 8px 16px;">Book
Ticket</button>
          </div>
        </div>
      </div>

      <div id="bookingModal" style="display:none; margin-top: 20px; border-top: 2px solid #eceff1; padding-top: 15px;">
        <h4 id="bookingTrainTitle"></h4>
        <input type="text" id="pName" placeholder="Passenger Full Name" style="width: 48%; display: inline-block;">
        <input type="number" id="pAge" placeholder="Age" style="width: 20%; display: inline-block;">
        <select id="pBerth" style="width: 28%; display: inline-block;">
          <option>Lower Berth</option><option>Middle Berth</option><option>Upper Berth</option><option>Side
Lower</option>
        </select><br><br>
        <button onclick="confirmTrainBooking()" style="width: 200px;">Generate PNR Ticket</button>
      </div>

      <div id="pnrDisplay"></div>

      <hr style="margin: 30px 0;">
      <h3>PNR Status Inquiry & Refund Calculation</h3>
      <div style="display: flex; gap: 10px;">
        <input type="text" id="checkPnr" placeholder="Enter 10-digit PNR Number" style="max-width: 300px;">
        <button onclick="lookupPNR()" style="width: 150px; background: #003366;">Check Status</button>
        <button onclick="cancelTrainTicket()" style="width: 180px; background: #d32f2f;">Cancel & Get Refund</button>
      </div>
      <div id="pnrStatusResult"></div>
    </div>

    <script>
      let pnrDatabase = {};

      function searchTrains() {
        document.getElementById('trainList').style.display = 'block';
      }

      function openBooking(tName) {
        document.getElementById('bookingModal').style.display = 'block';
        document.getElementById('bookingTrainTitle').innerText = 'Booking for: ' + tName;
      }

      function confirmTrainBooking() {
        let name = document.getElementById('pName').value.trim();
        let age = document.getElementById('pAge').value;
        let cls = document.getElementById('tClass').value;
        if(!name || !age) return alert('Enter passenger details');
        let fare = cls.includes('1250') ? 1250 : (cls.includes('840') ? 840 : 320);
        let pnr = Math.floor(1000000000 + Math.random() * 9000000000).toString();

        pnrDatabase[pnr] = { name, age, class: cls, fare, status: 'CONFIRMED (Coach B2, Berth 23)', date:
document.getElementById('jDate').value };

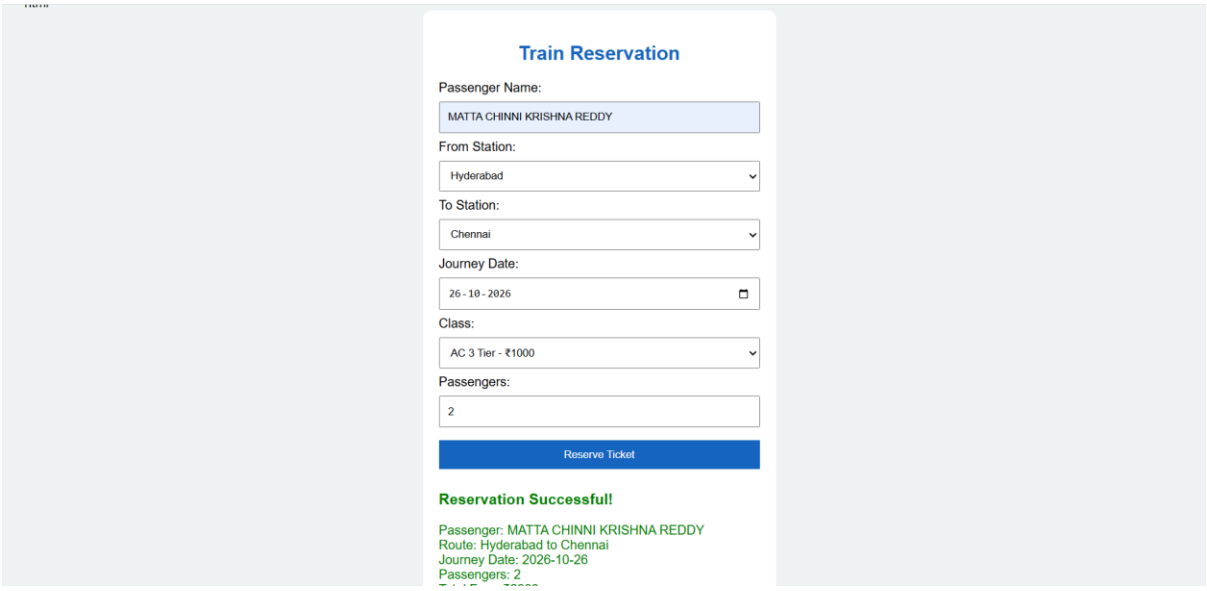
        document.getElementById('pnrDisplay').innerHTML = `
          <div class="pnr-box">
            <h3 style="color:#003366; margin-top:0;">E-Ticket Generated Successfully!</h3>
            <p><b>PNR Number:</b> <span style="font-size: 18px; color: #ff6600; font-weight: bold;">${pnr}</span></p>
```

```
            <p><b>Passenger:</b> ${name} (${age} yrs) | <b>Status:</b> ${pnrDatabase[pnr].status}</p>
            <p><b>Class:</b> ${cls} | <b>Fare Paid:</b> Rs. ${fare}</p>
        </div>`;
    }

    function lookupPNR() {
        let pnr = document.getElementById('checkPnr').value.trim();
        if(pnrDatabase[pnr]) {
            let rec = pnrDatabase[pnr];
            document.getElementById('pnrStatusResult').innerHTML = `<p style="color:#003366; font-weight:bold;">PNR ${pnr}: ${rec.name} | Class: ${rec.class} | Current Status: ${rec.status}</p>`;
        } else {
            document.getElementById('pnrStatusResult').innerHTML = `<p style="color:red;">Invalid or Unregistered PNR number.</p>`;
        }
    }

    function cancelTrainTicket() {
        let pnr = document.getElementById('checkPnr').value.trim();
        if(!pnrDatabase[pnr]) return alert('PNR not found');
        let fare = pnrDatabase[pnr].fare;
        let refundPct = 75; // Cancellation charge 25%, refund 75%
        let refundAmount = (fare * refundPct) / 100;
        delete pnrDatabase[pnr];
        document.getElementById('pnrStatusResult').innerHTML = `<div style="background:#ffebee; color:#c62828; padding:10px; border-radius:4px; font-weight:bold;">Ticket Cancelled! Refund of Rs. ${refundAmount} (${refundPct}% of Rs. ${fare}) initiated to customer account.</div>`;
    }
  </script>
</body>
</html>
```

## Output:




## RESULT:

***Thus, the online train reservation system was successfully built and tested, featuring multi-class train search, PNR issuance, live status enquiry, and automated cancellation refund calculation.***